`tmfem`: a validated two-dimensional thermo-mechanical finite-element solver written from scratch

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Abstract

`tmfem` is a C++17 finite-element solver for 1D and 2D transient heat conduction and diffusion, 2D linear elasticity, small-strain von Mises elastoplasticity and one-way thermo-mechanical coupling. It uses no FEM library. Every capability is tied to a validation test against an analytical solution, a recomputed textbook result or a reference simulation, with an explicit tolerance. The suite has 93 tests: 0 fail, and 1 is skipped (see below). They record 115 values for 32 reference rows; one more reference row (an ANSYS comparison) is skipped because its data were never provided. Observed orders of convergence match theory on four mesh levels: the T6 Kirsch plate converges at order 3.11 in the L^2 norm and 1.97 in the energy norm (levels 1–4, see Section 3.5), and it gives a stress concentration factor of 2.999 69 (exact: 3). A beam-heated block, explicitly a demonstration rather than a validation, illustrates the workflow used for beam-intercepting devices: at equal deposited energy, graphite has a 2.65 times larger temperature rise than copper yet carries 20 times less stress. `tmfem` is a learning and verification project. It does not replace ANSYS or Abaqus.

1 Context and goals

Beam-intercepting devices at CERN (targets, collimators, absorbers, dumps) are designed in two steps. A Monte Carlo transport code such as FLUKA [1] or Geant4 [2] computes where the beam deposits its energy. A finite-element code (in practice ANSYS) then computes the temperature field and the thermal stresses it causes, and these are checked against the strength of the material. `tmfem` rebuilds the second step from first principles, at small scale, as a way to understand and verify each ingredient: the weak forms, the element technology, time integration, return mapping plasticity and the thermal load.

The guiding rule is that no capability is claimed without a test that proves it. Each benchmark has a reference value and a tolerance in a reference table. Most rows were written before the code; the rows for 2D heat with source and convection and for the phase-4 convergence orders (P4-f to P4-h) were written in phase 4 together with their tests, as were the plane-strain targets of P4-a and P4-b. Every convergence claim is backed by an observed order measured on at least three refinements. The source material for the benchmarks (unpublished lab exercises by M. R. Kabiri, ENSAM Meknès, used with permission) contains several wrong values. Each reference was therefore recomputed by a script, and the disagreements are listed in Section 3.5. A test was never tuned to match a reference. When a result missed its tolerance, the cause was found, and any change to the test plan was approved and is disclosed (Section 3.5).

Scope. In scope: 1D and 2D transient heat and diffusion; 2D linear elasticity in plane stress and plane strain; small-strain J2 plasticity with isotropic hardening; one-way thermal-to-mechanical coupling; VTU output for ParaView. Out of scope: 3D, large deformation, dynamics and stress waves, temperature-dependent properties, contact, radiation and real particle-shower energy maps.

2 Methods

2.1 Heat conduction and diffusion

The heat equation $\rho c\,\dot{T} = \nabla\cdot(k\nabla T) + q$ is discretised with standard Galerkin finite elements. Boundary conditions are Dirichlet, prescribed flux, or convective $-k\,\partial T/\partial n = h\,(T - T_\infty(t))$, with $T_\infty(t)$ read from a time series and interpolated linearly. Diffusion uses the same code with $\rho c = 1$ and $k = D$. With capacity matrix $\boldsymbol{M}$ (consistent, or row-sum lumped in 1D), conductivity-plus-Robin matrix $\boldsymbol{A}$ and load $\boldsymbol{F}$, the θ-method reads

$$(\boldsymbol{M} + \theta\,\Delta t\,\boldsymbol{A})\,\boldsymbol{T}^{n+1} = (\boldsymbol{M} - (1-\theta)\,\Delta t\,\boldsymbol{A})\,\boldsymbol{T}^{n} + \Delta t\,\big[\theta\,\boldsymbol{F}^{n+1} + (1-\theta)\,\boldsymbol{F}^{n}\big], \tag{1}$$

with $\theta = 1$ (backward Euler, BE) or $\theta = 1/2$ (Crank–Nicolson, CN). Backward Euler is used when Δt is much larger than the diffusion time of an element. Crank–Nicolson is not L-stable and makes the stiff modes ring there [3]. Dirichlet values are imposed by elimination. The same integrator serves 1D (linear elements) and 2D (T3 and T6 triangles, volumetric source, convective edges and a convective face sink for thin plates).

2.2 Linear elasticity

Displacements are interpolated with three-node (T3) or six-node (T6) isoparametric triangles. On a curved boundary the T6 mid-side nodes lie on the curve. Gmsh [4] places them there, and this is required for optimal convergence on the hole of the Kirsch plate [5]. The stiffness is $\boldsymbol{K} = \sum_e \int_{\Omega_e} \boldsymbol{B}^T \boldsymbol{D}\,\boldsymbol{B}\,\mathrm{d}A$ in Voigt order $[xx, yy, xy]$ with engineering shear strain, and $\boldsymbol{D}$ is the plane-stress or plane-strain elasticity matrix. Quadrature uses one point for T3 and three for T6 stiffness. Nonlinear residuals use one point (T3) or the six-point degree-4 rule [6] (T6), and error norms collapsed Gauss rules (16 points for elasticity, 25 for heat). Tractions are integrated along element edges. Stresses are evaluated at quadrature points, or at any point inside the element that contains it (averaged over elements that share the point). Nodal averages are used only for plots and VTU output. Meshes are read from Gmsh 4.1 ASCII files, which the build generates from committed `.geo` files with Gmsh pinned to version 4.12.1, so that the same Gmsh build gives the same meshes everywhere.

2.3 Elastoplasticity

The material model is small-strain J2 plasticity with isotropic hardening $\sigma_y(\alpha)$, where α is the accumulated equivalent plastic strain. It is integrated by the backward-Euler radial return [7, 8]. Given the trial deviatoric stress $\boldsymbol{s}^{tr}$ and $q^{tr} = \sqrt{3/2}\,\|\boldsymbol{s}^{tr}\|$, the plastic multiplier solves the scalar equation $q^{tr} - 3G\,\Delta\gamma - \sigma_y(\alpha_n + \Delta\gamma) = 0$ by Newton's method. The consistent (algorithmic) tangent [9] is

$$\boldsymbol{C}^{alg} = \kappa\,\boldsymbol{1}\otimes\boldsymbol{1} + 2G\Big(1 - \frac{3G\Delta\gamma}{q^{tr}}\Big)\mathbb{I}_{dev} + 6G^2\Big(\frac{\Delta\gamma}{q^{tr}} - \frac{1}{3G+H}\Big)\,\bar{\boldsymbol{n}}\otimes\bar{\boldsymbol{n}}, \qquad \bar{\boldsymbol{n}} = \frac{\boldsymbol{s}^{tr}}{\|\boldsymbol{s}^{tr}\|}, \tag{2}$$

where κ is the bulk modulus and $H = \mathrm{d}\sigma_y/\mathrm{d}\alpha$. Two hardening laws sit behind one interface. The first is a piecewise-linear table built from the true tensile curve, with $\alpha = \varepsilon - \sigma/E$. The second is Ludwik's law [10] $\sigma_y = \sigma_{y0} + K\alpha^m$, fitted to the same data by least squares, with a covariance estimate and a leave-one-out jackknife for the uncertainty. Plane strain runs the 3D algorithm exactly on the four components $[xx, yy, zz, xy]$. In plane stress each quadrature point solves for ε_{zz} by a local Newton iteration so that $\sigma_{zz} = 0$. The tangent is then condensed, $D_{\alpha\beta} = C_{\alpha\beta} - C_{\alpha z}C_{z\beta}/C_{zz}$. The global equations are solved by Newton–Raphson with load stepping, and history variables are updated only from the converged state of the previous step. A material-point driver iterates the lateral strains until the lateral stresses vanish; this reproduces the uniaxial tensile test.

2.4 Thermo-mechanical coupling

The temperature field from the heat solver, on the same mesh, enters the mechanical problem as an eigenstrain $\alpha_{th}(T - T_{ref})$. Writing $\boldsymbol{m} = [1, 1, 0]^T$, the thermal load vector is

$$\boldsymbol{f}_{th} = \sum_e \int_{\Omega_e} \boldsymbol{B}^T\boldsymbol{m}\,\beta\,\Delta T_h\,\mathrm{d}A, \qquad \beta = \frac{E\alpha_{th}}{1-\nu}\ \text{(plane stress)}, \quad \beta = \frac{E\alpha_{th}}{1-2\nu}\ \text{(plane strain)}, \tag{3}$$

and stress recovery subtracts $\beta\,\Delta T_h\,\boldsymbol{m}$. In plane strain the out-of-plane stress is $\sigma_{zz} = \nu(\sigma_{xx} + \sigma_{yy}) - E\alpha_{th}\Delta T$. The case output once dropped the thermal term here, until an output test caught it. The coupling is one way and quasi-static: stress does not feed back into the temperature, and inertia is neglected.

2.5 Software and verification set-up

The code is about forty small C++17 translation units in `mesh`, `fem`, `thermal`, `mechanics`, `coupling` and `io`. It depends only on the standard library, Eigen 3 (sparse Cholesky LDL^T for symmetric systems, sparse LU otherwise), nlohmann/json for case files and Catch2 for tests. Cases are small JSON files. Each validation test writes the value it asserts to a CSV file next to its reference and tolerance. The README results table and Table 1 are generated from these files. Every other computed number in this report is generated from test or solver output by a script that also writes its source (`docs/PROVENANCE.md`). Inputs and reference values are quoted from the case files and `docs/REFERENCE_VALUES.md`. Continuous integration builds the code with warnings as errors and runs every test except the two slow elastoplastic plate tests on every push. The full suite is a manually started workflow.

3 Validation

Table 1 summarises the suite. For each reference row it shows the recorded quantity that uses the largest share of its tolerance; the README lists all 115 values with their test names. The worst two-sided row uses 80 % of its tolerance; the one-sided Newton-rate row (P3-d, scored as threshold over value) is at 85 %. The 2D orders (P2-f, P4-g, P4-h) are least-squares slopes of $\log e$ against $\log h$ over levels 1–4; level 0 is computed and printed but excluded from the fit (see Section 3.5). The 1D orders (P1-m) use all four levels.

3.1 Heat conduction and diffusion

Carburizing (P1-b) is solved on a semi-infinite domain truncated at more than $10\sqrt{Dt}$. The time for the carbon content to reach 0.8 wt% at 0.5 mm is 25 373.7 s, against 25 370.2 s from the error-function solution [11]. *A thin film on a half-space* (P1-h), started from the analytic profile at small $t_0 > 0$, conserves the dose to 3.7×10^{-12} relative, and its peak decays with slope $-0.499\,996$ (theory $-1/2$). The diffusion exercises P1-a to P1-g check closed-form results, and FE solutions where the exercise has a field, against recomputed values.

The steel plate (P1-i–P1-k), 5 mm thick and convective on both faces, is thermally thin: Biot number 0.001 83, time constant 549.5 s. Driven by a synthetic daily sol-air signal, the 1D FE inner-face temperature stays within 0.033 K of a lumped ODE solved with RK4 (0.026 K at half the time step). The difference is partly backward-Euler time error and partly the physical gap, of order Biot, between the face and the plate mean. The largest through-thickness difference is 0.0256 K, as predicted by the quasi-steady resistance network. The comparison with the source's ANSYS run (P1-l) is not done: the real input file and the ANSYS curve were never provided.

Convergence (P1-m, Fig. 1a). On four levels (three refinements) the observed temporal orders are 0.98 (BE) and 2.00 (CN) on an eigenmode problem, and 1.00 and 2.00 with a source, time-dependent Dirichlet data and Robin conditions. The spatial order is 2.00 with consistent and 2.01 with lumped capacity.

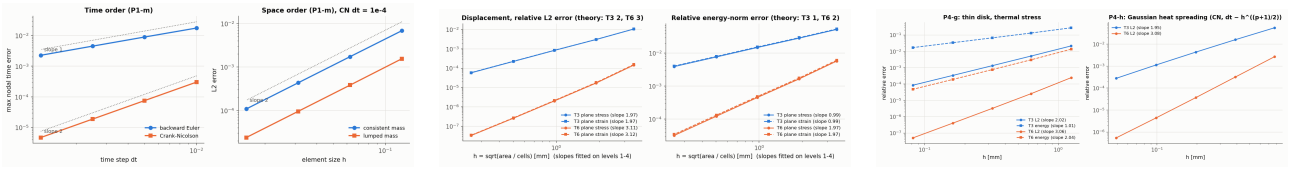
3.2 Linear elasticity

Patch test (P2-a): a linear displacement field is reproduced on a distorted mesh to 5.8×10^{-15} (T3) and 8×10^{-14} (T6) relative error, and the uniform bar (P2-b) to round-off.

Kirsch plate (P2-c–P2-f). A circular hole of radius a in an infinite plate under remote stress $S = 140$ MPa along y (Kirsch's solution, as given in [12]) is modelled as the quarter $[0, 5a]^2$ with symmetry conditions. The exact Kirsch tractions are applied on the outer edges, so the finite model

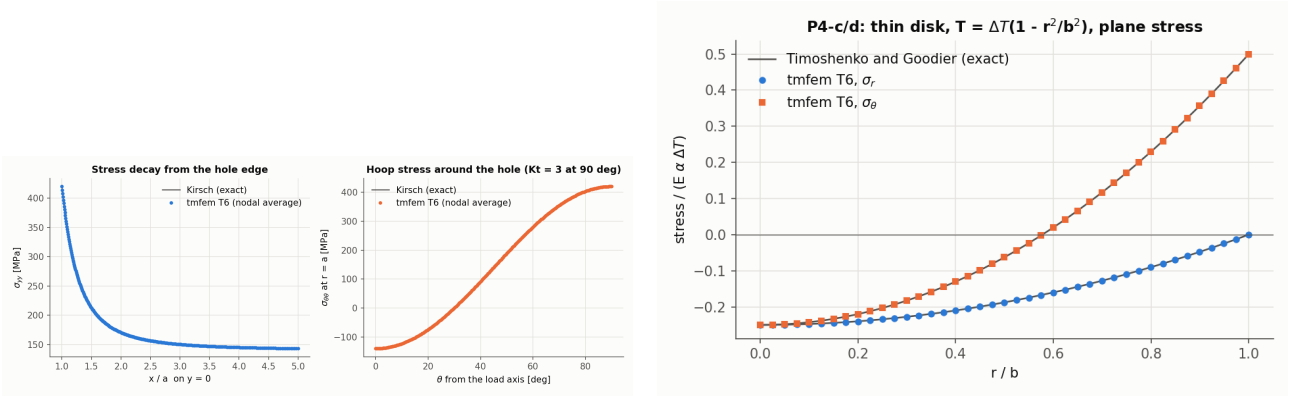
ID	Benchmark	Quantity (largest use of tolerance)	Reference	tmfem	Tolerance	Use
P1-a	Steady diffusion flux (ex. 1)	J finite element	2.4×10^{-9}	2.4×10^{-9}	1e-12 relative	0 %
P1-b	Carburizing, erf solution (ex. 2)	time finite element [s]	25 370	25 373.7	0.5%	3 %
P1-c	Equivalent time (ex. 3)	t at 500 C [h]	110.4	110.417	0.001 relative	15 %
P1-d	Arrhenius D(T) (ex. 4)	D(800 K) [m2/s]	2.14×10^{-15}	2.1396×10^{-15}	0.005 relative	4 %
P1-e	Boron predeposition dose (ex. 5)	dose formula [at/cm2]	$4.062 \times 10^{+15}$	$4.062\,17 \times 10^{+15}$	0.005 relative	1 %
P1-f	Thin-film surface concentration (ex. 5)	surface C at 1200 C [at/cm3]	$6.97 \times 10^{+19}$	$6.973\,82 \times 10^{+19}$	0.005 relative	11 %
P1-g	Arrhenius fit (ex. 5)	D(1300 C) [cm2/s]	1.74×10^{-11}	$1.741\,54 \times 10^{-11}$	0.005 relative	18 %
P1-h	Thin film on a half-space	max relative dose drift	0	$3.690\,59 \times 10^{-12}$	1e-10	4 %
P1-i	Steel plate parameters	Biot number	0.001 83	0.001 833 33	+/- 5e-6 (last quoted digit)	67 %
P1-j	Steel plate, FE vs lumped ODE	max abs inner face FE - lumped [K]	0	0.033 065 2	0.05 K	66 %
P1-k	Steel plate, through-thickness difference	max through-thickness difference [K]	0.025 534 8	0.025 583 2	5%	4 %
P1-m	1D heat convergence orders	time order BE	1	0.983 31	+/- 0.15	11 %
P2-a	Patch test, distorted mesh	T6 max relative error (u, sigma)	0	$7.969\,39 \times 10^{-14}$	1e-10 relative	0 %
P2-b	Uniform bar in tension	T6 max rel error of end displacement	0	$1.229\,86 \times 10^{-13}$	1e-8 relative	0 %
P2-c	Kirsch plate, stress concentration	Kt finest T6	3	2.999 69	1%	1 %
P2-d	Kirsch plate, r/a = 2 and 4	max abs stress error, 20 rows x 3 components [MPa]	0	0.004 411 35	2 MPa	0 %
P2-e	Kirsch plate, r/a = 1.5	max abs stress error, 10 rows x 3 components [MPa]	0	0.011 764 4	2 MPa	1 %
P2-f	Kirsch convergence orders	T6 plane strain L2 order (levels 1-4)	3	3.120 65	+/- 0.15	80 %
P3-a	Ludwik fit to the true tensile curve	log-log K [MPa] (sigma_y fixed)	548	547.608	0.5%	14 %
P3-b	Material point, uniaxial, tabulated hardening	eps_yy at 457.24 MPa	0.031 58	0.031 581	0.5%	1 %
P3-c	Material point, Ludwik along the true curve	max |driver - data| / s, LSQ Ludwik, 7 points	0	1.5942	+/- 2 s	80 %
P3-d	Newton convergence rate	Newton rate, last qualifying triple, plane-strain bar, Ludwik	2	2.120 18	>= 1.8	85 %
P3-e	Plane-strain bar FE vs material point	max relative difference FE vs driver (T3, T6, tabulated, Ludwik)	0	$1.115\,96 \times 10^{-12}$	1e-6 relative	0 %
P3-f	Elastoplastic hole plate, S = 140 MPa (report only)	report only (20 values), see §3.3				
P4-a	Bar held in x, uniform dT	T6 plane strain max rel error of sigma_xx	0	1.0396×10^{-13}	1e-8 relative	0 %
P4-b	Free body, uniform dT	T6 plane strain max stress over beta dT	0	$5.265\,08 \times 10^{-14}$	1e-10 relative	0 %
P4-c	Thin disk, parabolic T: sigma_r	sigma_r over E alpha dT at r/b = 0.0	−0.25	−0.249 959	1%	2 %
P4-d	Thin disk, parabolic T: sigma_theta	sigma_theta over E alpha dT at r/b = 0.5	−0.0625	−0.062 477 4	1%	4 %
P4-e	2D Gaussian heat spreading	T3 max rel variance error	0	0.000 349 37	0.5%	7 %
P4-f	2D heat with source and convection	(iii) amplification T6 theta 0.5	0	$4.244\,82 \times 10^{-11}$	1e-10 relative	42 %
P4-g	Thin disk convergence orders	T6 plane stress L2 order (levels 1-4)	3	3.056 64	+/- 0.15	38 %
P4-h	2D Gaussian heat convergence orders	T6 L2 order (levels 1-4)	3	3.078 25	+/- 0.15	52 %
P1-l	Steel plate, real ttab vs ANSYS	skipped: references/data/ttab.csv and the ANSYS curve are not provided				

Table 1: Validation summary, one row per reference row of `docs/REFERENCE_VALUES.md`. "Use" is |error| divided by the tolerance, in the tolerance's own units (threshold over value for the Newton rate). Generated from the test records by `scripts/report_numbers.py`.



(a) 1D heat (P1-m) (b) Kirsch plate (P2-f) (c) Thin disk, Gaussian (P4-g, h)

Figure 1: Observed convergence. Each claimed order is asserted within ±0.15 on levels 1–4.



(a) Kirsch: FE against exact stresses (b) Thin disk: FE against Timoshenko–Goodier

Figure 2: Point stresses against closed-form solutions (T6, finest mesh).

represents the infinite plate exactly and the FE solution converges to the analytic field. On the finest T6 mesh the stress concentration factor is 2.999 69 (T3: 2.973), and the 60 tabulated stress components at $r/a = 2$ and 4 agree within 0.0044 MPa (tolerance 2 MPa), the 30 at $r/a = 1.5$ within 0.012 MPa. The observed orders (Fig. 1b) are 1.97 and 0.99 for T3 (L^2, energy; theory 2 and 1) and 3.11 and 1.97 for T6 (theory 3 and 2), in plane stress. The plane-strain orders differ from these by at most 0.012.

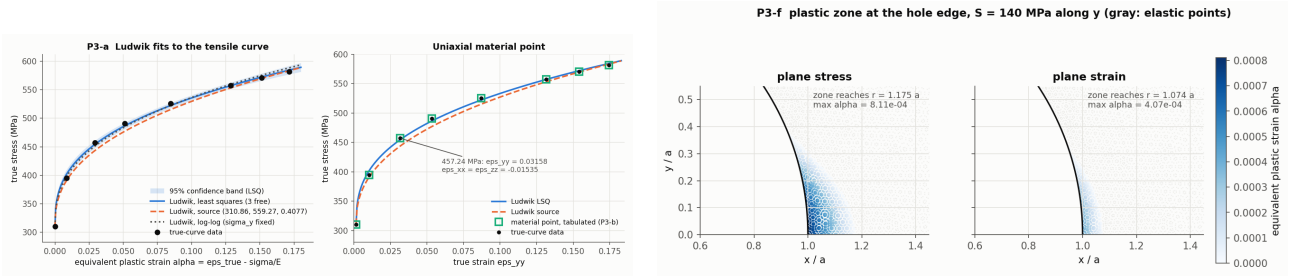
3.3 Elastoplasticity

Parameter identification (P3-a, Fig. 3a). With σ_y fixed at the yield point, a log-log fit reproduces the recomputed reference ($m = 0.385$, $K = 548$ MPa). The deliverable is a three-parameter least-squares fit on stress: $\sigma_{y0} = 309.1$ MPa, $K = 520.9$ MPa, $m = 0.362$, with residual standard deviation 4.58 MPa. The source gives 310.86 MPa, 559.27 MPa and 0.4077. The jackknife standard errors are much larger than the covariance ones, because no data lie between $\alpha = 0$ (the yield point) and $\alpha = 0.0086$, so the yield point alone fixes the knee. This is a limit of the eight-point data set, not of the solver.

Material point (P3-b). With tabulated hardening at $\sigma = 457.24$ MPa the uniaxial driver gives $\varepsilon_{yy} = 0.031\,581$ and $\varepsilon_{xx} = \varepsilon_{zz} = -0.015\,35$, against the Abaqus values 0.03158 and −0.01535. The check is nearly tautological for ε_{yy}, since 457.24 MPa is a data point. The lateral strain is the informative part: $-\nu\sigma/E$ elastic plus $-\alpha/2$ plastic. With Ludwik hardening (P3-c) the driver follows the data within 1.59 residual standard deviations. This band is a weak test: with eight points and three parameters no residual can exceed $\sqrt{5}\,s = 2.24\,s$.

Newton convergence (P3-d). The observed rate in the last iterations of a load step is 2.12 (plane strain) and 2.62 (plane stress) on the uniform FE bar with Ludwik hardening. This shows that the tangent is consistent along the loading path. The full global tangent is also compared with central differences of the internal force on a non-uniform plastic Kirsch state. *FE bar against the driver* (P3-e): a uniform plane-strain bar meshed with T3 or T6 reproduces the material point with $\varepsilon_{zz} = 0$ to 1.1×10^{-12} relative at every load step.

Hole plate at $S = 140$ MPa (P3-f, report only, Fig. 3b). The elastic peak of 420 MPa exceeds the yield stress, so a plastic zone forms. In plane stress, the thin plate of the source's Abaqus model, the T6 level-4 solution gives an edge strain $\varepsilon_{yy} = 0.002\,356$, a peak equivalent plastic strain of 0.000 815 and $\sigma_{yy} = 318.7$ MPa at the quadrature point nearest the edge. The plastic zone extends $0.174\,a$ from



(a) Tensile data, fits and material point

(b) Plastic zone at the hole, $S = 140$ MPa

Figure 3: Elastoplasticity: parameter identification and the hole plate.

the hole. Neuber's rule $\sigma\varepsilon = (K_t S)^2/E$ [13], solved on the same hardening curve, gives $\varepsilon = 0.002\,647$, $\alpha = 0.001\,098$ and $\sigma = 321.3$ MPa. The FE strain differs by $-11.0\,\%$ and the plastic strain by $-25.7\,\%$, while the stresses differ by $-0.8\,\%$. Neuber usually overestimates notch strains, and it is reported as an estimate, not a target. In plane strain the notch root is biaxial ($\sigma_{zz} = 124$ MPa near the edge), and the uniaxial rule does not apply.

3.4 Thermo-mechanical coupling and 2D heat

A bar held in x under a uniform temperature rise (P4-a) and a free body (P4-b) are reproduced to round-off in both plane states. They check β, the thermal load and stress recovery, including σ_{zz} in plane strain. For a thin disk with $T = \Delta T(1 - r^2/b^2)$ (P4-c, P4-d, Fig. 2b) the Timoshenko–Goodier solution [12] gives $\sigma_r/(E\alpha\Delta T) = -0.25$, -0.1875, 0 at $r/b = 0$, 0.5, 1. The finest T6 mesh gives $-0.249\,96$, $-0.187\,48$ and 2.2×10^{-5}, and for σ_θ it gives $-0.249\,96$, $-0.062\,477$, $0.500\,014$ (exact -0.25, -0.0625, 0.5). The disk converges at orders 2.02 and 1.01 (T3) and 3.06 and 2.04 (T6).

In 2D heat, an initial Gaussian spreading in a large domain (P4-e) keeps its variance on $s_0^2 + 2\alpha t$ within $0.000\,35$ (T3) relative, and it conserves energy to 5.8×10^{-14}. On straight T6 the variance identity holds almost by construction, so the L^2 error of the whole profile is the real accuracy test (P4-h). It converges at order 1.95 (T3) and 3.08 (T6). These are combined space–time orders along $\Delta t \propto h^{(p+1)/2}$ with CN; at level 2 the time part is 2 % (T3) and 50 % (T6) of the error. The source and convection terms (P4-f) pass three exactness checks: a steady slab with a uniform source and Robin faces, a uniform plate cooled through its faces against the discrete amplification factor, and a discrete energy balance. The last is an identity of the θ-scheme, so the same test also checks $\boldsymbol{A}\,(T_\infty \mathbf{1}) = \boldsymbol{F}$ on a curved Robin rim with a face sink (5.5×10^{-17}), which a wrong load would fail.

3.5 Errors found in the source, and disclosed changes

Table 2 lists the source values that the recomputation does not confirm. The test targets are the recomputed values.

Five changes to the test plan were made after a first run missed its target. Each was approved before it was made, and `docs/VALIDATION.md` records the numbers from before the change. (i) P1-k: the original bound is physically infeasible and was replaced by the quasi-steady bound. (ii) P2-f: the T6 L^2 slope over levels 0–3 was outside 3 ± 0.15. The pairwise orders were falling toward 3, so level 0 is pre-asymptotic. A finer level 4 was added, and the fit now uses levels 1–4. The pass depends on dropping level 0, and this is stated. (iii) P3-d: on the Kirsch plate Newton reached round-off too fast for the pre-fixed rate criterion, so the rate is asserted on the bar, where the active set is fixed by construction. (iv) P4-e: the variance test moved to the finest mesh after a constant $O(h^2)$ interpolation offset missed 0.5% on level 2. (v) P4-f(iii): the plate is posed in excess temperature with $k = 0.4\,\mathrm{W/(m\,K)}$ instead of 400 (the target G^n does not depend on k), because cancellation and a conduction-dominated solve at $T_\infty = 300\,\mathrm{K}$ exceeded 10^{-10} in floating point. In addition, the P4-c tolerance was clarified in phase 4 to an absolute 0.0025 where the target is zero (σ_r at $r = b$). The CI runner is pinned to Ubuntu 24.04, not "latest", because the build requires Gmsh 4.12.1.

Row	Quantity	Source says	Recomputed	Cause
P1-a	steady flux J	2.4×10^{-12} kg/(m^2 s)	2.4×10^{-9} kg/(m^2 s)	mm not converted
P1-b	time to 0.8 wt%	7.025 h	7.047 h	erf table interpolation
P1-f	surface concentration	2.8×10^{19} cm^{-3}	6.97×10^{19} cm^{-3}	value matches t = 36 min
P1-d	D(1000 K), D(800 K)	2.04×10^{-13} m^2/s, 2.11×10^{-15} m^2/s	2.067×10^{-13} m^2/s, 2.140×10^{-15} m^2/s	not identified
P1-g	D_0, D(1300 °C)	3.2 m^2/s, 5.86×10^{-11} m^2/s	3.11 cm^2/s, 1.74×10^{-11} cm^2/s	unit slip (D_0); D(1300 °C) not identified
P1-k	through-thickness ΔT	< 0.01 K	0.0255 K	infeasible for a 48 K swing
P2-e	σ_{rr} at $r = 1.5a$, $\theta = 0$	2100 MPa	25.93 MPa	wrong table entry
P3-a	Ludwik m, K	0.4077, 559.27 MPa	0.385, 548 MPa	different point choice

Table 2: Source values not confirmed by recomputation (`references/verify_reference_values.py`).

4 Demonstration: a beam-heated block

This section is a demonstration, not a validation. A Gaussian beam of rms size s deposits $E_l = 1$ kJ per metre of path in a pulse of 10 µs into a long block of 40 × 40 mm cross-section. The deposit is $q = E_l/(2\pi s^2\tau)\exp(-r^2/2s^2)$, a stand-in for a FLUKA or Geant4 energy map, which would vary along the beam and would not be Gaussian. The pulse is adiabatic, and afterwards the outer edges cool by convection ($h = 10$ W/(m^2 K)). The cross-section is solved in plane strain on T6 meshes refined to $s/10$ near the axis, for $s = 0.5$, 1, 2 and 4 mm. A long free block is really in generalised plane strain. The difference is a uniform σ_{zz} shift of 0.064 % to 4.1 % of the peak $|\sigma_{zz}|$ for copper (graphite: up to 5 %), and twice that fraction in the centre von Mises stress. The materials are annealed OFE copper [14] and isotropic graphite (SGL R4550 [15], with $\nu = 0.2$ and c from NIST-JANAF [16] assumed, because the data sheet does not give them). Every property and its source is recorded in `cases/beam_block/README.md`.

Results (Fig. 4). The FE centre values agree with closed-form infinite-plane estimates within 0.5 % (temperature, which diffuses during the pulse) and 4.4 % (von Mises, because the section is finite). At $s = 1$ mm copper heats by 46.2 K and carries 74.3 MPa von Mises at the centre, while graphite heats by 123 K and carries only 3.7 MPa. Graphite's temperature rise is larger because its ρc is 2.65 times lower. It is less stressed because the stress scales with the thermal-shock figure of merit $E\alpha/((1-\nu)\rho c)$, which is 0.047 Pa m^3/J for graphite against 0.93 Pa m^3/J for copper. The stress is mostly compressive and out of plane: the cold surroundings prevent the hot core from expanding. The peak stress falls as $1/s^2$. Annealed copper (yield 69 MPa) exceeds yield for $s \le 1$ mm: the ratio of von Mises stress to yield is 4.29 at 0.5 mm and 1.08 at 1 mm. Graphite reaches only 0.054 of its flexural strength and 0.23 of its compressive strength, even at 0.5 mm. The tensile check is optimistic, because flexural strength overestimates tensile strength. This illustrates one reason graphite is used in dumps and absorbers, its thermal-shock resistance. Its other main advantage, a lower energy density for the same beam thanks to its low density and atomic number, is outside this demo, which fixes the deposited energy. After the pulse the stresses relax as heat diffuses over the spot: for copper at $s = 1$ mm the centre stress is 31 % of its peak at 10 ms and 4.4 % at 100 ms, long before convection removes the energy.

Validity. The results are elastic with constant properties. Above yield, copper would need thermoplasticity. With constant c, graphite at $s = 0.5$ mm reaches 782 K, but its specific heat at 800 K is 2.3 times the room-temperature value. Integrating the JANAF $c(T)$ for the same deposit gives a rise of 319 K (to 612 K) instead of 491 K, so that row overestimates the temperature rise by 54 %.

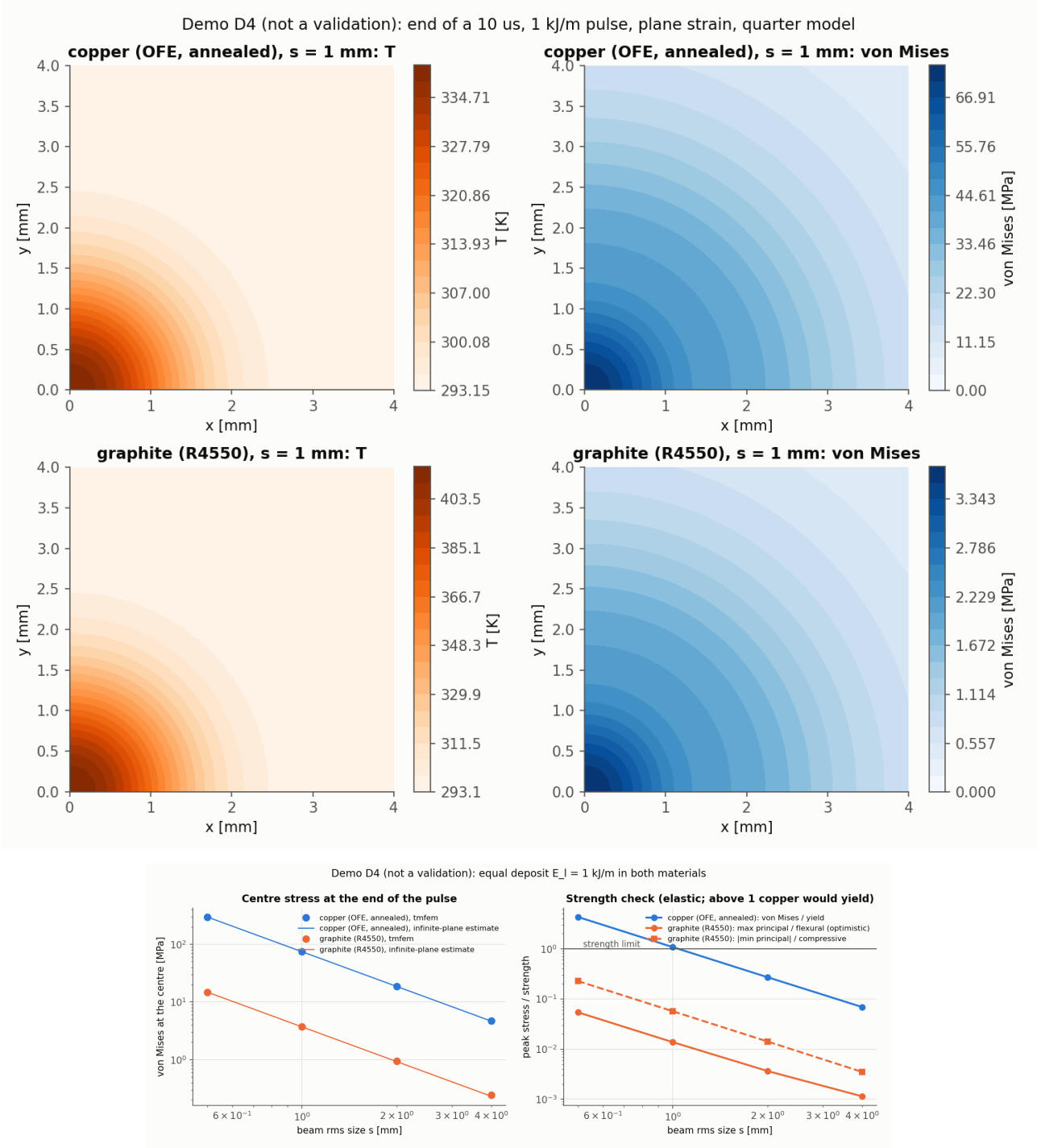


Figure 4: Demo D4 (not a validation). Top: temperature and von Mises stress at the end of the pulse. Bottom: centre stress and strength ratios against beam size.

5 Limitations

- **Out of scope by design:** 3D, large deformation, dynamics and stress waves, temperature-dependent properties, contact, radiation, real energy maps. A 10 µs pulse is long compared with the acoustic transit time of a millimetre spot, but stress waves are not modelled.
- **Elements and materials:** triangles only (T3, T6); one isotropic material per run; unit thickness; no body force. J2 plasticity with isotropic hardening only, so there is no Bauschinger effect. The model is rate-independent and isothermal. The tabulated law has slope jumps, so quadratic Newton convergence is asserted with Ludwik hardening only.
- **Benchmarks:** the Kirsch plate uses infinite-plate tractions, not the finite plate of the source's Abaqus model. The elastoplastic hole plate keeps elastic Kirsch tractions on its outer boundary, which is an approximation once a plastic zone exists, and is reported only. The steel-plate case is validated against a lumped model with a synthetic input, not against ANSYS (P1-I is skipped). The fit uncertainty describes eight digitised points and model misfit, not material scatter.
- **Time integration:** fixed time steps without error control. In 2D only consistent capacity is available, because row-sum lumping gives zero capacity to T6 corner nodes.
- **Reproducibility:** meshes come from Gmsh 4.12.1. Another version would give slightly different meshes and numbers, so the build refuses other versions.
- **Demo:** plane strain instead of generalised plane strain, Gaussian deposit, elastic constant-property analysis, and assumed graphite ν and c (Section 4).

6 Conclusion and outlook

`tmfem` implements, from scratch, the chain of methods behind a thermal-stress analysis of a beam-heated component. Each link is tied to a test against an independent answer, including observed convergence orders for every discretisation. The process found several errors in the reference material, and every deviation from the original test plan is disclosed. Possible next steps, not attempted here, are stress-intensity factors by the domain J-integral for a cracked strip, OpenMP assembly, temperature-dependent properties with thermoplasticity for the demo, and generalised plane strain.

Credits. Benchmark cases adapted from M. R. Kabiri, ENSAM Meknès lab materials (unpublished), with permission. Analytical solutions from the cited textbooks.

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A Reproducing the numbers

Requirements: a C++17 compiler, CMake $\geq$ 3.16, Eigen 3 and Gmsh 4.12.1 (Ubuntu 24.04: `apt install libeigen3-dev gmsh`). Catch2 and nlohmann/json are fetched at pinned versions.

```
cmake -S . -B build -DCMAKE_BUILD_TYPE=Release
cmake --build build -j
ctest --test-dir build -V > results/ctest_full.log 2>&1
./build/src/tmfem cases/beam_block/copper.json    # demo, section 4
./build/src/tmfem cases/beam_block/graphite.json
python3 scripts/build_results_table.py            # README table
python3 scripts/report_numbers.py                 # numbers, provenance
cd report && latexmk -pdf main.tex
```

`docs/PROVENANCE.md` lists every macro used in this report with its value, source file and the test or command that produced it.